DATERA EDU: Project Proposal

1. Overview

DATERA EDU is a web system that listens to a recorded lesson and tells the teacher, the student and the parent which topics the student understood and which need more work.

Full title: Development of an intelligent system for monitoring students' knowledge acquisition using speech recognition and large language model technologies for analysing teacher-student dialogue (Russian: Разработка интеллектуальной системы контроля усвоения знаний обучающихся с использованием технологий распознавания речи и больших языковых моделей для анализа диалога преподавателя и ученика).

Supervisor: Khaimuldin Nursultan Gazizovich, senior lecturer, Astana IT University.

Team member	**Role**	**What they do**
David Khamitskiy	Project manager, ML/AI	Requirements and priorities; audio processing and speech recognition; the language-model analysis and its fine-tuning
Tairkhan Yermek	Backend developer	Server (Go), database, task queue, file storage, login, connection to the ML service
Yermakhan Khamitov	Frontend developer	Web interface for three roles, reports, knowledge map, audio player

2. The problem and who it is for

In tutoring and language schools, what a student understood in a lesson is rarely written down anywhere.

Teachers recall earlier lessons from memory. Parents learn about progress mostly from the teacher’s short comments. Recordings are almost never re-watched, because watching takes as long as the lesson. When a student changes teacher, the history of weak topics is lost.

Research shows that classroom speech can be analysed automatically, but the published work looks at teaching quality or at single steps of the process. No system was found that turns a lesson recording into a topic-by-topic picture of what a student has learned and explains it to the teacher, the student and the parent (see the literature review and the comparison of existing systems).

The statements about current teacher and parent practice are working assumptions. They were checked in five short interviews, and the results are in the Evidence of Relevance document. Two assumptions were not covered by the interviews and will be checked in the pilot.

Who will use it

- Teachers and independent tutors: they see what each student did and did not understand, without re-listening to the lesson.
- Online schools and learning centres: one place for the history of every student, even when teachers change.
- Students: a short list of what to repeat (tests on weak topics come in the second stage).
- Parents: a brief report in plain language.

The pilot covers English lessons (including lessons explained in Russian) and Russian-language lessons, 2-3 subjects in total.

3. Aim and objectives

Aim. To build and test a web system that, from a lesson recording and the teacher's own course materials, shows which topics a student has mastered and which not, and gives the teacher, the student and the parent a report on it.

How it works

1. The teacher uploads course materials (PDF, Word, slides, text). The system splits them into topics, and the teacher confirms the list.
2. The teacher uploads the recording of a one-to-one lesson.
3. The system turns the speech into text, separates teacher and student, and a language model checks the student's answers against the materials, topic by topic.
4. Reports and a knowledge map appear. The teacher can confirm or change any rating, and the last word is always theirs.

Objectives

1. Review the literature and compare existing systems.
2. Build the audio pipeline: noise reduction, recognition of English, Russian and mixed speech, separation of teacher and student voices.
3. Build topic extraction from the materials and answer checking, with time-coded evidence from the lesson.

4. Design a mastery index and a colour status for each topic, and compare them with the teacher's own judgement.
5. Build the web system: three roles, material upload, reports and a knowledge map. In the second stage: tests, weekly summaries and notifications.
6. Collect the teachers' confirmations and corrections as training examples, fine-tune an open language model, and compare it with the base and cloud models.
7. Run a pilot with 2-3 teachers and measure speech recognition, speaker separation, analysis quality and processing time.

4. Deliverables

A working MVP is planned by March 2027; after it come a second stage and the results of the study.

- **MVP (March 2027).** The teacher uploads materials and a one-to-one lesson recording. Within 30 minutes (for a 60-minute lesson) the system produces three reports (teacher, student, parent) and a knowledge map with colour statuses. The teacher can confirm or change each rating.
- **Second stage (March-April).** Tests on weak topics (questions can be generated from the teacher's materials), weekly summaries and notifications.
- **Measured quality.** On at least 20 real recordings: word error rate (WER), speaker-separation error (DER), correct teacher detection in at least 95 % of recordings. On a control set of 5-10 lessons rated by a teacher: topic statuses agree with the teacher in at least 70 % of cases. At least 95 % of language-model answers are valid.
- **Mastery index method.** A documented method, chosen by comparing several variants against the teacher's judgement.
- **Fine-tuning study.** At least 300 training examples (goal: 500), taken from the teachers' confirmations and corrections. A comparison of the base open model, the fine-tuned model and a cloud model, and a decision on whether the fine-tuned one can replace the cloud model (it must be within 5 percentage points of it).
- **Pilot and handover.** Pilot results with 2-3 teachers, source code, API description and the explanatory note.

5. Scope

The project builds a realistic first version and leaves the harder parts for later.

In scope	**Out of scope (future work)**

English, Russian and mixed English-Russian lessons	Kazakh and Kazakh-Russian speech: recognition is much less reliable (see the literature review). It is only measured on 3-5 recordings.
One-to-one lessons	Group lessons: they need voices matched to students
Upload of the teacher's materials (PDF, DOCX, PPTX, text) and splitting them into topics	Automatic recognition of students by voice
Upload of audio and video files up to 90 minutes	Recording in the browser, Zoom and Google Meet integration
Web interface for the organisation (teacher), the student and the parent	Mobile app, Telegram or push notifications
Mastery index calculated by rules from the analysis; the teacher can override it	Risk prediction by machine learning, teacher-effectiveness analytics
Fine-tuning of the analysis model (open data first, then teachers' corrections)	Fine-tuning of the noise-reduction and speech-recognition models
Second stage: tests, weekly summaries, notifications	Lesson plans suggested by the language model

The system gives the teacher a suggestion; it does not grade students. Audio is stored for 30 days, and recordings of minors are processed only with a parent's consent. Personal data stays on servers in Kazakhstan, and only anonymised text is sent to an external language model.

6. Technical approach

The project joins a web system, a chain of machine-learning models and a research part.

- **Audio and speech (Python).** Preparation with ffmpeg, noise reduction (DeepFilterNet, Demucs or RNNoise, chosen by measured error), speech recognition (faster-whisper large-v3), speaker separation (pyannote.audio).
- **Language-model analysis.** The model splits the materials into topics and checks the student's answers against them. A cloud model is used in the MVP. A thin adapter allows replacing it later with an open model fine-tuned within the project.
- **Backend (Go).** Roles and consent, uploads, a task queue (RabbitMQ), PostgreSQL, file storage (MinIO). The mastery index and statuses are calculated by fixed rules, not by the language model, so results can be reproduced.
- **Frontend (React, TypeScript).** Three role-based cabinets, knowledge map, audio player linked to time codes, and the teacher's confirm-or-change buttons.
- **Research part.** Noise-reduction comparison, recognition quality on mixed speech, how recognition errors affect the analysis, the choice of index method, and the fine-tuning experiment.

Main risks

Risk	**Mitigation**
Recognition is worse on real lessons than on test data	Measure on the first recordings in November, compare models, mark low-accuracy lessons
Too little data for fine-tuning	Trial on open datasets first; examples accumulate from teachers' corrections from the first lesson; describe the result honestly
Fine-tuned model is not as good as the cloud model	The cloud model stays; the comparison is a research result
No GPU server yet	Research on open data in Colab or Kaggle; real lessons only on a server in Kazakhstan

The project manager is overloaded	Monthly review of tasks; the backend developer takes the ML integration
The MVP is late	The MVP core (materials, recording, analysis, reports) is protected; the second stage is cut first

Each risk has a detailed fallback in the technical specification (section 9.1).

Timeline

When	**Milestone**
Nov 2026	Requirements approved, development starts, test recordings collected
Nov 2026 - Feb 2027	Backend, interface, analysis, reports and knowledge map
Late winter - early spring 2027	Fine-tuning of the analysis model
Mar 2027	MVP on ready-made models
Mar - Apr 2027	Pilot with 2-3 teachers; second stage (tests, weekly summaries, notifications)
Apr - May 2027	Explanatory note